

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and optimized **medical device components** by translating design inputs into engineering specifications, material selection, and process parameters; conducted **design qualification and validation** activities including **verification and validation protocols** and **test method development** for **pre-clinical and clinical studies**, delivering **15+ design iterations** with **100% regulatory compliance**.
- Developed and executed **process development activities** using **Design of Experiments (DOE)** and statistical methods to optimize key process inputs and outputs; supported **equipment development**, vendor selection, and **new process equipment** qualification, reducing process variability by **32%** and cycle time by **18%**.
- Prepared comprehensive technical documentation including **work instructions, standard operating procedures (SOPs), raw material specifications, and engineering drawings**; authored **verification and validation reports** and **test method validation protocols** with full traceability and regulatory audit readiness across **40+ device components**.
- Developed **engineering test methods** and **clinically relevant simulated use test methods** in collaboration with R&D; technicians; conducted **lab bench testing, design evaluations, and competitor evaluations**, executing **verification and validation testing** across **8 distinct test scenarios** with **99.2% repeatability**.
- Performed hands-on **lab bench testing and device optimization** activities including **engineering evaluations and development unit testing**; led cross-functional collaboration with **Quality, Operations, and Regulatory Affairs** to support design reviews, change-control management, and post-implementation validation, accelerating **3 device launches** with **zero critical findings**.
- Trained **R&D; technicians and operations teams** on new design methodologies, process development workflows, and **test method validation** procedures; developed technical training materials and onboarding documentation, enabling **12+ technicians** to independently execute **qualification and validation activities** within **6 weeks**.

SKILLS

Design & Engineering – component design, engineering specifications, materials selection, design inputs and outputs, design qualification, design validation, clinical use scenarios, international standards.

Process Development & Optimization – Design of Experiments (DOE), statistical methods, process inputs and outputs, process optimization, equipment development, vendor selection, process tooling, new process equipment.

Testing & Validation – lab bench testing, engineering evaluations, competitor evaluations, development unit testing, verification and validation testing, design qualification testing, clinically relevant simulated use testing, test method validation.

Technical Documentation – work instructions, standard operating procedures (SOPs), engineering drawings, raw material specifications, technical reports, verification and validation protocols, validation reports, design documentation.

Test Method Development – engineering test method development, clinically relevant test method design, test method validation, test method documentation, pre-clinical testing, clinical study support.

Cross-Functional Collaboration & Training – R&D; support, Quality assurance, Operations coordination, Regulatory Affairs liaison, technician training, process development training, change-control management, UAT coordination.

Software & Data Tools – Python, SQL, Power BI, data analysis, statistical analysis, R, technical reporting, documentation systems.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Medical Device Component Optimization & Validation Platform (HCL Technologies)

- Led **end-to-end design and process development** for a **critical catheter assembly component**, translating clinical requirements into **engineering specifications, material selections, and manufacturing processes**; executed **Design of Experiments (DOE)** across **6 process parameters**, reducing defect rates from **8.3% to 1.2%** and enabling commercial-scale production.
- Designed and validated **5 distinct test methods** including **clinically relevant simulated use testing, tensile strength verification, and biocompatibility validation protocols**; authored **verification and validation protocols and reports** with **statistical confidence >99%**, supporting regulatory submissions across **US FDA, EU MDR, and ISO 13485** requirements.
- Developed comprehensive technical documentation suite comprising **engineering drawings, work instructions, SOPs, and raw material specifications** for **12 manufacturing sub-processes**; coordinated **qualification and validation activities** with Operations and Quality teams, achieving **zero compliance findings** during regulatory inspection.
- Conducted **lab bench testing and engineering evaluations** including competitor benchmarking and development unit testing across **200+ test units**; identified **3 design optimizations** that improved product durability by **28%** and reduced manufacturing complexity by **15%**.
- Trained **18 R&D; technicians and 8 Operations personnel** on new **test method validation procedures, process development methodologies, and design qualification workflows**; delivered **4 comprehensive training modules** with hands-on lab exercises, enabling independent execution of validation activities and reducing time-to-production by **6 weeks**.

SWIFT Payments Operations Dashboard & Process Validation (HCL Technologies)

- Engineered and optimized a **production-grade payments operations dashboard** for a major banking client, implementing **ISO 20022 message flow tracking, SEPA compliance validation, and cross-border payment monitoring**; reduced **payment processing SLA violations** by **34%** and improved transaction-flow visibility for **daily volumes exceeding 250K messages**.
- Developed **test protocols and validation procedures** for **SWIFT MT/MX message formats** and **payments processing workflows**; executed **pre-production testing and UAT coordination** with banking operations teams and regulatory liaisons, ensuring **100% GDPR and Data Protection Act compliance**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)